



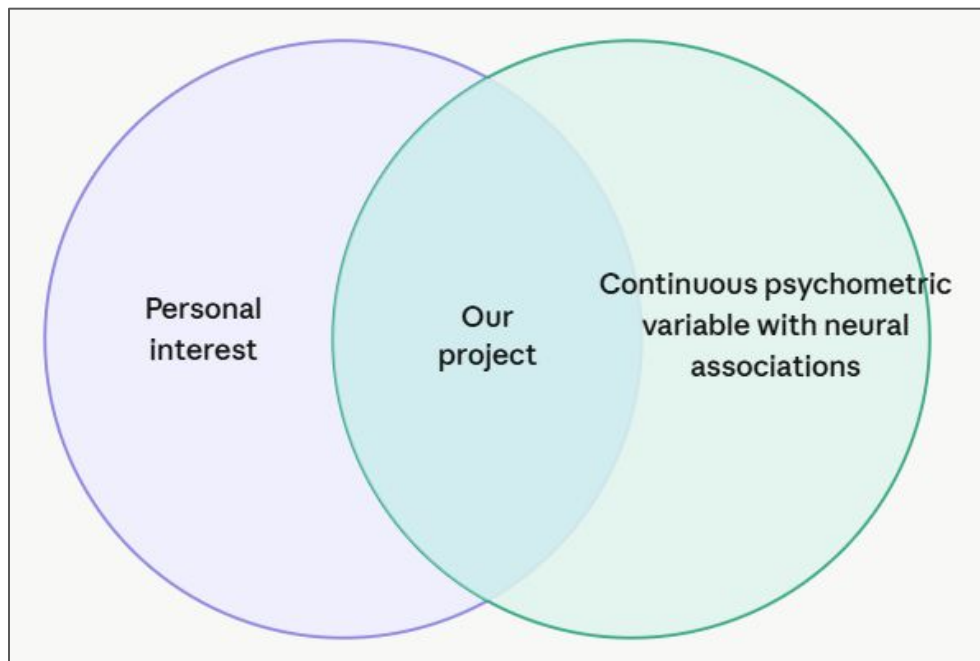
Multidimensional Approach to Examining Aggression Vulnerability: Psychosocial Factors and Structural Variations in the Amygdala and Prefrontal Cortex

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Background

- Why Aggression?



Background

- Majority of aggression research rely on **self-report measures** and **behavioural paradigms** (Roberton et al., 2012)
 - Studied in silos from biological/neural explanations for aggression
 - reactive aggression that is rapid and emotionally driven, where immediate action is prioritized over long-term consequential thinking is not able to be fully explained by self-report measures

Background

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 - Studied in silos from biological/neural explanations for aggression
 - reactive aggression that is rapid and emotionally driven, where immediate action is prioritized over long-term consequential thinking is not able to be fully explained by self-report measures
- Existing neuroimaging research (Rosell & Siever, 2015; Siever, 2008) suggests that aggression is associated with:
 - Poor emotional regulation ↔ emotion-processing systems (e.g., amygdala)
 - low inhibitory control ↔ reward-processing systems
 - maladaptive decision-making ↔ executive-control systems (e.g., prefrontal cortex)
 - **BUT** relevant brain systems are often studied independently

Research Gap

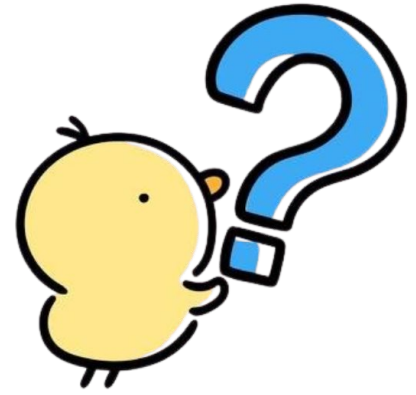


Examine the relationship between **psychosocial factors** and the **structural variations** in the relevant neural systems (**amygdala** and **prefrontal cortex**) and whether these structural characteristics are associated with **vulnerability to aggressive behaviours**.

Literature Review

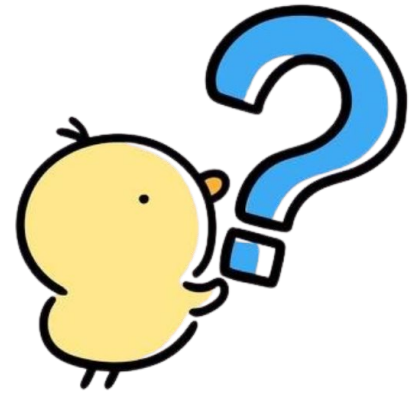
Psychosocial Factors (6)	Relationship with Aggression	Citations
Stress	↑ Stress ↑ Aggression	Anderson & Bushman, 2002
Self-efficacy	↓ Self-efficacy ↑ Aggression	Mesurado et al., 2018; Roberton et al., 2012
Self-regulation	↓ Self-regulation ↑ Aggression	Roberton et al., 2012; Smith et al., 2025
Social Relationships	↓ Support, ↑ Conflict, ↑ Rejection, ↑ Aggression	Heilbron & Prinstein, 2008
Executive Function	↓ Inhibition, ↑ Aggression	Choy et al., 2025; Pawliczek et al., 2013
Big 5 Personality traits	↓ Agreeableness, ↓ Conscientiousness, ↑ Aggression	Jones et al., 2011

Research Question



Do **psychosocial factors** (stress, self-efficacy, self-regulation, social relationships, and personality traits) predict **structural variations** in **prefrontal cortex and amygdala** in **aggression vulnerability**?

Hypothesis



H1 (Risk factors): ↑ **stress** and **composite aggression** (anger, hostility, perceived rejection) will be associated with ↓ **PFC** volume/thickness and/or ↑ **amygdala** volume.

H2 (Protective factors): ↑ **self-efficacy**, **executive function**, **agreeableness**, and **conscientiousness** will be associated with ↑ **PFC** volume/thickness and/or ↓ **amygdala** volume.

Methods

Dataset

Human Connectome Project Young Adult S1200 Release ($n = 1206$)

- Age: **22-35** years old
- Data collected between **2012 -2015**



HCP variables of interest

Demographics	Gender Age Range
Psychosocial factors	(1) NEOFAC_A, (2) NEOFAC_C (Agreeableness, Conscientiousness) SelfEff_Unadj (Self-efficacy) PercStress_Unadj (Perceived stress) AngAffect_Unadj (Anger-Affect) AngHostil_Unadj (Anger-Hostility) AngAggr_Unadj (Anger-Physical Aggression) Flanker_Unadj (Flanker Inhibitory Control and Attention Test) Flanker_AgeAdj (Flanker Inhibitory Control and Attention Test: Age-Adjusted) PercHostil_Unadj (Perceived Hostility) PercReject_Unadj (Perceived Rejection)
Brain structures	FS_L_Amygdala_Vol, FS_R_Amygdala_Vol FS_L_Medialorbitofrontal_GrayVol, FS_R_Medialorbitofrontal_GrayVol FS_L_Rostralanteriorcingulate_GrayVol, FS_R_Rostralanteriorcingulate_GrayVol FS_L_Medialorbitofrontal_Thck, FS_R_Medialorbitofrontal_Thck FS_InterCranial_Vol

Methods

Analysis Plan

- Neuroimaging data: structural MRI
 - Compare the structural variations (volume, grey matter volume and cortical thickness) between PFC and amygdala
 - Brain ROI: Prefrontal cortex (cognitive control) and amygdala (emotion processing)
 - Data analysis methods: Pearson correlations + FDR correction
-

Results

Correlations between brain structure (amygdala, mOFC, rACC volumes/thickness, normalized for head size) and psychosocial measures (stress, self-efficacy, anger, hostility, perceived rejection, Flanker inhibition, agreeableness, conscientiousness) were examined across ~1100 HCP subjects.

All correlations were very weak ($|r| \leq 0.08$).

Largest associations observed:

- rACC gray matter volume $\leftrightarrow$ Anger-Aggression ($r \approx 0.08$)
- Amygdala volume $\leftrightarrow$ Flanker inhibitory control ($r \approx -0.08$)
- rACC gray matter volume $\leftrightarrow$ Agreeableness ($r \approx -0.07$)

After Benjamini-Hochberg FDR correction (44 comparisons), [insert: X out of 44 pairs survived at $q < 0.05$] — fill this in from your `corr_results['significant_fdr'].sum()` output.

Even where statistically significant, $r \approx 0.06$ – 0.08 explains less than 1% of variance — indicating a weak/null brain-behavior pattern.

Discussion

Key findings:

- Structural variations were not strong predictors
 - Structural variation in the amygdala, mOFC, and rACC showed minimal relationships with aggression-related traits
- Aggression may not be strongly reflected in regional brain structure
 - Structural measures alone may be insufficient to capture aggression vulnerability

Future research should examine integrated biopsychosocial and functional neural mechanisms (e.g., functional neural processes, brain network interactions).

THANK YOU

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why is this important - background/intro (problem statement/ research question/aim)

-literature review

-theory

-methods

-dataset we are gg to use (HCP)

-variables

-neuroimaging technique

-analysis plan

-potential expected findings

Variables:

- Stress & self efficacy (emotion regulation) W
- self regulation (emotion regulation) W
- Executive Function/Inhibition (Flanker Inhibitory Control and Attention Task) (impulsivity) M
- Social Relations (environmental factor) W
- Big 5 (personality) M

Brain parts:

- Prefrontal Cortex (neuro - cognitive control) W&M
- Amygdala (neuro - emotions) W&M

Demographic:

age

gender

education

Literature review

- Prefrontal Cortex (Rosell & Siever, 2015)
 - Supports cognitive control, impulse inhibition, planning, and top-down regulation of emotional responses
 - ↓ prefrontal control ↑ difficulty to inhibit aggressive responses under anger or provocation
- Amygdala (Siever, 2008)
 - involved in emotional salience, threat processing, fear, anger, and responses to provocative or negative stimuli
 - impulsive aggression involves heightened limbic reactivity

Research gap: Lack of studies examining how psychosocial factors relate to functional connectivity of the relevant neural systems (amygdala and prefrontal cortex) and how these relevant neural systems are interconnected

Literature Review

- Stress
 - **Stress as risk:** negative affect, anger, and physiological arousal ↑ impulsive aggression
- Self efficacy (Mesurado et al., 2018; Robertson et al., 2012)
 - **Self efficacy as protective:** ability to manage difficult emotion can better regulate anger and avoid aggressive responses to provocation
- Self regulation (Robertson et al., 2012; Smith et al., 2025)
 - Key to aggression: involves controlling impulses, managing anger, and selecting non-aggressive responses under provocation
 - under- and over-regulated emotion ↑ likelihood of aggression
- Social Relations (Heilbron & Prinstein, 2008)
 - Aggression is shaped by interpersonal context, support, rejection, and social conflict
 - relational difficulties (e.g., peer rejection, exclusion, and conflict) associated with aggression and antisocial behaviour

Literature review

- Executive Function/Inhibition (Flanker Inhibitory Control and Attention Task)
 - High trait aggression was not associated with reduced inhibitory control on the Flanker task (Krämer et al., 2011)
 - ↓ inhibition ↑ aggression (Choy et al., 2025; Pawliczek et al., 2013)
- Big 5 (Jones et al., 2011)
 - ↑ Aggression is related to ↓ Agreeableness and ↓ Conscientiousness, with smaller but meaningful links to ↑ Neuroticism.

Research Question and Hypothesis

RQ: Does functional connectivity between the amygdala and prefrontal cortex predict vulnerability to aggression?

Proposed RQ revision for clarity because our present rq doesnt mention how we are using the variables as factors of aggression: Do psychosocial factors (stress, self-efficacy, self-regulation, social relationships, and personality traits) predict prefrontal cortex–amygdala functional connectivity in aggression vulnerability?

H1: high stress = weaker PFC-amygdala functional connectivity

H2: low self efficacy = weaker PFC-amygdala functional connectivity

H3: low self regulation = weaker PFC-amygdala functional connectivity

H4: low social relationship = weaker PFC-amygdala functional connectivity

H5: low exec function = weaker PFC-amygdala functional connectivity

H6: low agreeableness = weaker PFC-amygdala functional connectivity

H7: low conscientiousness = weaker PFC-amygdala functional connectivity

How brain activity changes based on proxy data...

Each of our variables are factors of aggression...

Proposed methods

Dataset: Human Connectome Project Young Adult S1200 Release ($n = 1206$)

- Age: 22-35 years old
- Data collected between 2012 -2015

Analysis Plan

- Neuroimaging technique: resting state fMRI
- Brain areas: Prefrontal (cognitive control) and amygdala (emotion processing)
- Correlations/Regressions will be conducted

Data Analysis: Linear and logistic regression

Correlation

Once we establish the relationship we can explore the moderators and the relationship

Do the data analysis on R? SPSS?

Variables of Interest

	PercStress_Unadj
Age - restricted - propose dont include	SelfEff_Unadj
Gender	AngAffect_Unadj
SSAGA_Educ	AngHostil_Unadj
	AngAggr_Unadj
FS_L/R_Amygdala_Vol	Flanker_Unadj
rfMRI_REST1_LR_Atlas_MSMAII.dtseries.nii	Flanker_AgeAdj
rfMRI_REST1_RL_Atlas_MSMAII.dtseries.nii	PercHostil_Unadj (proxy)
rfMRI_REST2_LR_Atlas_MSMAII.dtseries.nii	PercReject_Unadj (proxy)
rfMRI_REST2_RL_Atlas_MSMAII.dtseries.nii	NEOFAC_N/E/O/A/C
	rs-fMRI connectivity